



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO PONNSANKAR AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

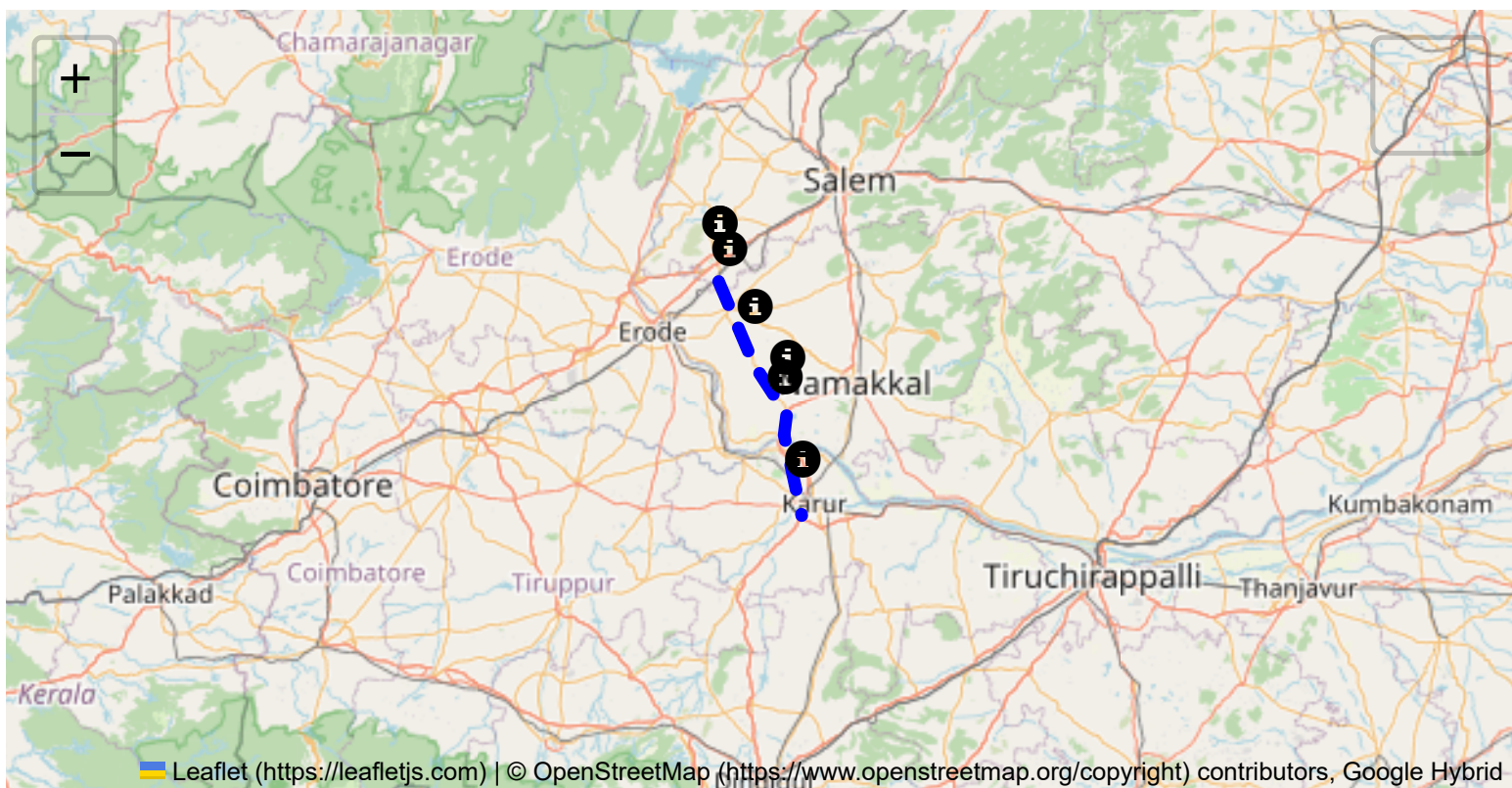
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

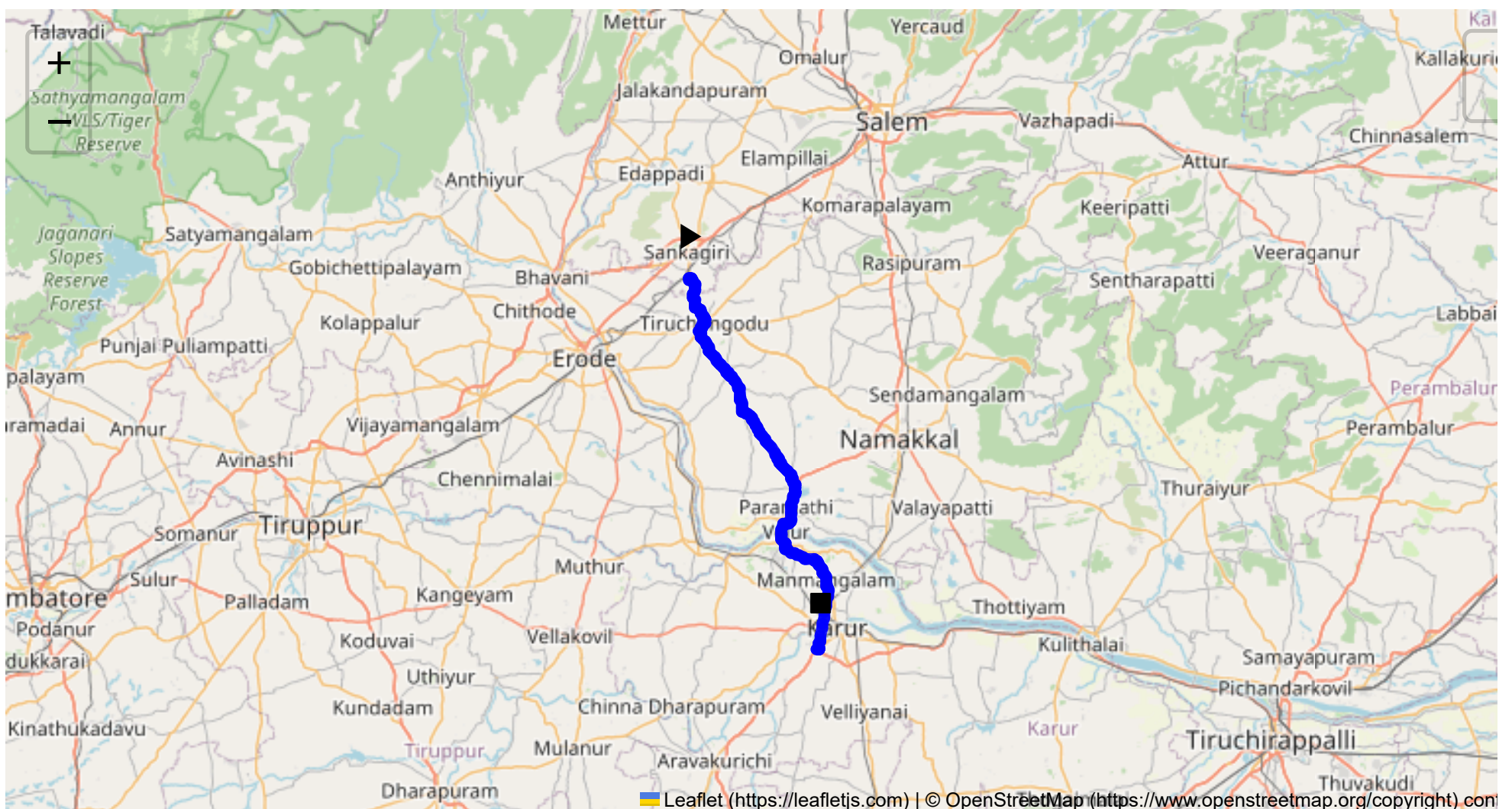
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 68.84 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (10.93411, 78.054802)



Welcome to the Journey Risk Management Study

Route Overview

The route spans approximately 68.84 kilometers, passing through a mix of rural and semi-urban areas in Tamil Nadu. It mainly follows local roads and state highways, connecting points such as Sangagiri, Puthur, Tiruchengode, Musalnaickenpalayam, Paramathi, Velur, and Karattuppalaiyam. This route traditionally serves as a connector between agricultural areas and more urbanized regions.

Weather Conditions and Hazards

- **Typical Weather:** The region generally experiences hot and humid conditions, with temperatures rising during the summer months (March to June). The monsoon season, from July to September, brings significant rainfall.
- **Potential Weather Hazards:** During monsoons, heavy rainfall may lead to waterlogging and reduced visibility, increasing the risk of accidents. It's crucial to monitor weather forecasts and plan travel accordingly.

Traffic Patterns

- **Peak Hours:** Most traffic congestion happens between 8:00-10:00 AM and 5:00-7:00 PM, corresponding with local commuting patterns.
- **Congestion-Prone Areas:** Urban sections, particularly around Tiruchengode and Velur, may experience slower movement. Highway intersections can also be bottlenecks.

Road Quality and Infrastructure

- **General Assessment:** The road quality is variable; highways are generally well-maintained, while some rural roads may have potholes or uneven surfaces.
- **Infrastructure Gaps:** Limited signage in rural areas may pose navigation challenges. Bridges and culverts should be approached with caution, particularly after heavy rains.

Alternative Routes for Emergencies

- **Primary Alternative:** In case of blockage, drivers may reroute via the NH44 and SH95 roads, though this may increase travel time. Local roads connecting smaller towns can serve as secondary detours.
- **Backup Options:** Familiarize with parallel routes to each checkpoint, as informed by local mapping resources.

Local Regulations

- **Hazardous Material Transport:** Transport of hazardous materials is regulated, requiring appropriate documentation and adherence to safety standards. Night travel may be restricted in some regions; check local regulations.

Historical Incidents

- **Reported Incidents:** There are sporadic reports of heavy vehicle accidents, generally attributed to road conditions and driver error. Due diligence in vehicle maintenance and adherence to speed limits is advised.

Environmental Considerations

- **Sensitive Areas:** Agricultural lands dominate the area; spills could severely impact soil and water quality. Extra care should be taken near water bodies to prevent contamination.
- **Conservation Zones:** Minimal; yet any signage indicating wildlife crossings should be respected.

Communication Coverage

- **Network Performance:** While main areas are covered by cellular networks, dead zones may exist in rural stretches, potentially challenging communication.
- **Improvement Strategy:** Pre-download maps and ensure alternative contact methods are considered.

Emergency Response Times

- **Urban Areas:** Faster response times, approximately 20-30 minutes.
- **Rural or Remote Areas:** Response could take up to an hour or more due to fewer emergency stations.

Overall Risk Assessment

The route presents moderate risks primarily related to weather, traffic congestion, and variable road conditions. Regularly updated route knowledge, weather monitoring, and keeping an archive of emergency contact numbers will help mitigate potential issues. Proper adherence to transport regulations and environmental care are essential to conduct a safe journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43961, 77.87341	15 KM/Hr
4	Turn	High	11.44022, 77.87546	15 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16755, 78.01568	30 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Turn	High	11.14480, 78.02291	15 KM/Hr
10	Turn	High	10.92955, 78.05373	15 KM/Hr
11	Turn	High	10.93406, 78.05512	15 KM/Hr
12	Turn	High	10.93420, 78.05503	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
6	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.92958, 78.05452



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44022, 77.87546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16755, 78.01568



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

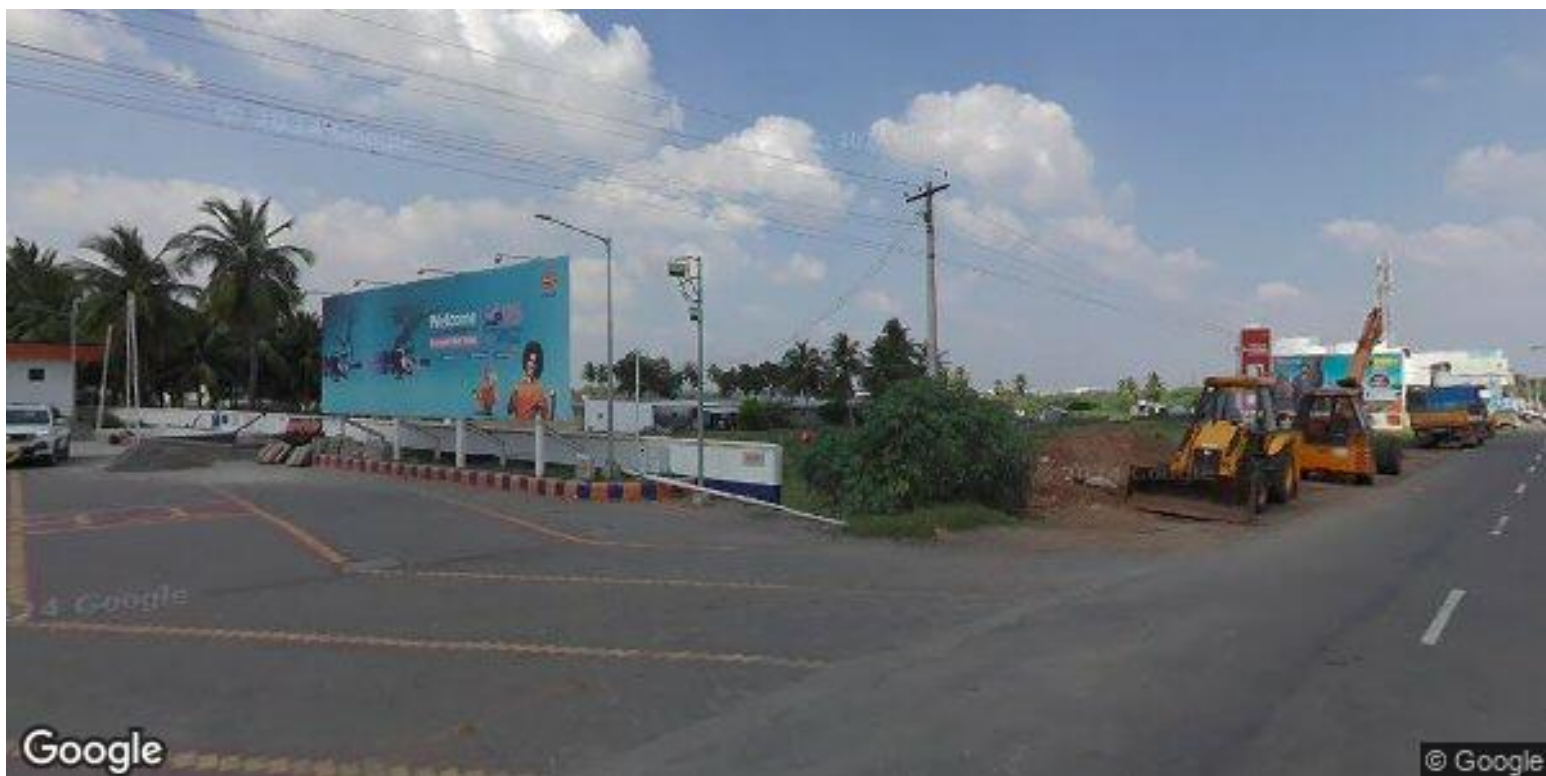
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.92955, 78.05373



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.93406, 78.05512



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.93420, 78.05503
